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BTEC

Mark Scheme (Results)

June 2023

Pearson BTEC Nationals
In Applied Science/Forensic and Criminal Investigation (31617H/1B)
Unit 1: Principles and Applications of Science I
Biology

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Unit 1: Principles and Applications of Science I Biology

General marking guidance

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark grids should be applied positively. Learners must be rewarded for what they have shown they can do rather than be penalised for omissions.
- Examiners should mark according to the mark grid, not according to their perception of where the grade boundaries may lie.
- All marks on the mark grid should be used appropriately.
- All the marks on the mark grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the mark grid.
- Where judgement is required, a mark grid will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the mark grid to a learner's response, a senior examiner should be consulted.

Specific marking guidance

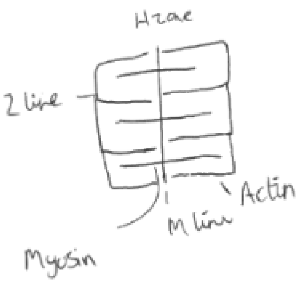
The mark grids have been designed to assess learners' work holistically.

Rows in the grids identify the assessment focus/outcome being targeted. When using a mark grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.

- Marks will be awarded towards the top or bottom of that band depending on how they have evidenced each of the descriptor bullet points.

Question Number	Answer	Additional Guidance	Mark
1 (a)	● tissue ◆ (organ) system	accept example of a tissue, e.g. muscle accept any example of an organ system, e.g. digestive system	2
1 (b)(i)	Award **two** marks for the correct answer. If the answer is incorrect, award **one** mark for each correct stage of the calculation: substitution (1) 40 ÷ 10 evaluation (1) (x) 4		2
1 (b)(ii)	B 0.2 µm		1
		Total	5 marks

Question Number	Answer	Additional Guidance	Mark
2 (a)	C nucleus		1
2 (b)	Award **one** mark for the following: myofibril accept any other valid response	accept muscle fibril, sarcostyle reject myofibre muscle cell sarcoplasm sarcolemma	1
2 (c)	Award **one** mark for each of the following: Z lines closer together (1) more overlap of the actin and myosin (1) 		2

2 (d)	Award **one** mark for an identification point and **one** mark for each expansion point, up to a maximum of **three** marks. Any one of the following could be an identification point, depending on how the learner shapes their response. (fast twitch muscle) can use anaerobic respiration (1) (fast twitch muscle) uses the (high levels of stored) glycogen (1) so that many glucose molecules are partially broken down (1) giving ATP quickly (1) (so) contract faster/ high force in short period of time (1) fatigues quickly so can't be used for endurance (1)	ORA allow energy for ATP Ignore references to lactic acid build up	3
		Total	7 marks

Question Number	Answer	Additional Guidance	Mark
3 (a)	B – cell wall		1
3 (b)(i)	Award **two** marks for the correct answer. If the answer is incorrect, award **one** mark for each correct stage of the calculation: substitution (1) π x 12^2 evaluation (1) 452.16 (mm^2)	award 2 marks for answers rounding to 452	2

3 (b)(ii)	Award **one** mark for an identification point and **one** mark for each expansion point, up to a maximum of **three** marks. Any of the following could be an identification point or an expansion point depending on how the learner shapes their response. (Gram-positive bacteria) have thick peptidoglycan/cell wall (1) penicillin only affects dividing cells (1) inhibits cell wall synthesis (1) new cells take in too much water (1) (leads to) bursting (1) Accept any other valid response.	allow reference to preventing multiplication or reproduction ignore breaks down cell wall	3
		Total	6 marks

Question Number	Answer	Additional Guidance	Mark
4 (a)	Award **one** mark for the following: depolarisation	allow description of depolarisation	1
4 (b)	−50 mV (1)	accept values between −45 and −55 reject any positive number	1

4 (c)	Award **one** mark for an identification point and up to **three** marks for each linked expansion point, up to a maximum of **four** marks. Any of the following may be an identification or an expansion point, depending on how the learner shapes their response. Na^+ and K^+ {pumps/proteins/carriers} in membrane (1) $3Na^+$ out and $2K^+$ in (1) by active transport/using ATP (1) K^+ ions <u>diffuse</u> out of the cell (1) (because) membrane more permeable to K^+ than Na^+ (1) proteins with –ve charge/Cl^- inside axon (1) so inside of membrane is negative (with respect to outside) (1)	less permeable/ impermeable to Na^+ ORA Allow reference to a negative value between -65 and -70mV for the inside of the membrane	4
		Total	6 marks

Question Number	Indicative content
5	Answers will be credited according to the learner’s demonstration of knowledge and understanding of the material, using the indicative content and levels descriptors below. The indicative content that follows is not prescriptive. Answers may cover some or all of the indicative content but learners should be rewarded for other relevant answers. **hypertension** can be caused by a number of factors, e.g. high salt diet, genetics, age, alcohol, high blood cholesterol/LDL, obesity, diabetes, alcohol, drugs, lack of exercise can be affected/raised or lowered by lifestyle or medication **causes smooth endothelial damage by** high pressure triggers inflammatory response macrophages/white blood cells move to site of damage ingest LDLs in blood foam cell formation (foam cells with) LDL and calcium ions sink under the damaged lining / into wall leads to formation of fatty streaks/atheroma/atherosclerosis narrows lumen of artery makes walls less elastic/weakens walls walls get thicker/ more smooth muscle added restricts blood flow increases blood pressure further mechanical damage to lining of arteries

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	**Mark**	**Descriptor**
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates adequate knowledge of scientific facts/concepts with generalised comments made. • Generic statements may be presented rather than linkages being made so that lines of reasoning are unsupported or partially supported. • The discussion shows some structure and coherence.
Level 2	3–4	• Demonstrates good knowledge and understanding by selecting and applying some relevant scientific knowledge facts/concepts to provide the discussion being presented. • Lines of argument mostly supported through the application of relevant evidence. • The discussion shows a structure which is mostly clear, coherent and logical.
Level 3	5–6	• Demonstrates comprehensive knowledge and understanding by selecting and applying relevant knowledge of scientific facts/concepts to provide the discussion being presented. • Line(s) of argument consistently supported throughout by sustained application of relevant evidence. • The discussion shows a well-developed structure which is clear, coherent and logical.

Llywodraeth Cynulliad Cymru
Welsh Assembly Government

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